

Certified Defenses

Lecture 5 — fixture deck for the span pipeline

Randomized smoothing

- Add Gaussian noise at prediction time
 - the noise level is chosen at training time
 - not at certification time
- Take a majority vote over the noisy copies
- The vote margin becomes a radius
 - 1) bigger margin, bigger certificate

The certificate

Noise scale σ^2 controls the trade-off.

$$\|x' - x\|_{\infty} \leq \varepsilon$$

Radius uses Φ^{-1} of the vote probability.

Two views

Empirical

Attack the model and report what survives. Cheap to run, easy to break later.

Certified

Prove a radius no attack can cross. Expensive, smaller numbers, but honest.

Alpha comes first on the page.

Bravo comes second on the page.

Charlie comes third on the page.

Delta comes fourth on the page.

Summary

Certified defenses trade accuracy for a guarantee that holds against every attack inside the radius, so the numbers look small.